Exp No:15 Experiment to understand the data preprocessing in Data Science

AIM:

To understand the data preprocessing in Data Science

Description:

Understand the importance of Data preprocessing in data science.

CODE:

```
import numpy as np
import pandas as pd
df = pd.read_csv("/content/customer_purchase_dataset.csv")
df
df.info()
df.City.mode()
df.City.mode()[0]
type(df.City.mode())
df.City.fillna(df.City.mode()[0], inplace=True)
df.Age.fillna(df.Age.median(), inplace=True)
df.Income.fillna(round(df.Income.mean()), inplace=True)
df
pd.get_dummies(df.City)
updated_dataset = pd.concat(
  [pd.get_dummies(df.City), df.iloc[:, [1, 2, 4, 5]]],
  axis=1
)
updated_dataset
updated_dataset.info()
updated:_dataset.Purchased.replace(['No', 'Yes'], [0, 1], inplace=True)
```

updated_dataset

OUTPUT:

	Customer_ID	Gender	Age	City	Income	Purchased
0	301	Male	22.0	Chennai	25000.0	No
1	302	Female	25.0	Bangalore	40000.0	Yes
2	303	Male	28.0	Chennai	52000.0	Yes
3	304	Female	24.0	Hyderabad	NaN	No
4	305	Male	30.0	Bangalore	65000.0	Yes
5	306	Female	27.0	Chennai	48000.0	Yes
6	307	Male	NaN	Hyderabad	35000.0	No
7	308	Female	29.0	Bangalore	NaN	Yes
8	309	Male	32.0	Chennai	72000.0	Yes
9	310	Female	26.0	Hyderabad	42000.0	No

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
0	Customer_ID	10 non-null	int64
1	Gender	10 non-null	object
2	Age	9 non-null	float64
3	City	10 non-null	object
4	Income	8 non-null	float64
5	Purchased	10 non-null	object

dtypes: float64(2), int64(1), object(3)

memory usage: 612.0+ bytes

	City
0	Chennai

dtype: object

	Customer_ID	Gender	Age	City	Income	Purchased
0	301	Male	22.0	Chennai	25000.0	No
1	302	Female	25.0	Bangalore	40000.0	Yes
2	303	Male	28.0	Chennai	52000.0	Yes
3	304	Female	24.0	Hyderabad	47375.0	No
4	305	Male	30.0	Bangalore	65000.0	Yes
5	306	Female	27.0	Chennai	48000.0	Yes
6	307	Male	27.0	Hyderabad	35000.0	No
7	308	Female	29.0	Bangalore	47375.0	Yes
8	309	Male	32.0	Chennai	72000.0	Yes
9	310	Female	26.0	Hyderabad	42000.0	No

	Bangalore	Chennai	Hyderabad
0	False	True	False
1	True	False	False
2	False	True	False
3	False	False	True
4	True	False	False
5	False	True	False

	Bangalore	Chennai	Hyderabad
6	False	False	True
7	True	False	False
8	False	True	False
9	False	False	True

	Bangalore	Chennai	Hyderabad	Gender	Age	Income	Purchased
0	False	True	False	Male	22.0	25000.0	No
1	True	False	False	Female	25.0	40000.0	Yes
2	False	True	False	Male	28.0	52000.0	Yes
3	False	False	True	Female	24.0	47375.0	No
4	True	False	False	Male	30.0	65000.0	Yes
5	False	True	False	Female	27.0	48000.0	Yes
6	False	False	True	Male	27.0	35000.0	No
7	True	False	False	Female	29.0	47375.0	Yes
8	False	True	False	Male	32.0	72000.0	Yes
9	False	False	True	Female	26.0	42000.0	No

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 7 columns):

Column Non-Null Count Dtype

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0 Bangalore 10 non-null bool

1 Chennai 10 non-null bool

2 Hyderabad 10 non-null bool

3 Gender 10 non-null object

4 Age 10 non-null float64

5 Income 10 non-null float64

6 Purchased 10 non-null object

dtypes: bool(3), float64(2), object(2)

memory usage: 482.0+ bytes

	Bangalore	Chennai	Hyderabad	Gender	Age	Income	Purchased
0	False	True	False	Male	22.0	25000.0	0
1	True	False	False	Female	25.0	40000.0	1
2	False	True	False	Male	28.0	52000.0	1
3	False	False	True	Female	24.0	47375.0	0
4	True	False	False	Male	30.0	65000.0	1
5	False	True	False	Female	27.0	48000.0	1
6	False	False	True	Male	27.0	35000.0	0
7	True	False	False	Female	29.0	47375.0	1
8	False	True	False	Male	32.0	72000.0	1
9	False	False	True	Female	26.0	42000.0	0

RESULT:

The given dataset was successfully preprocessed by handling missing values and preparing it for further analaysis.